

Recommended PET Configuration

Comparison with Gregor's Implementation

INTERIM INVENTORY, rev. 2 — the selection is not yet made

Publication Agent B (PET)

18 September 2026

This deck inventories differences. It does not select a configuration.
The matched comparison that would select one is specified on slides 16–19 and in
MATCHED_COMPARISON_PROPOSAL-20260918.md (rev. 2). Nothing is launched.

Slide 3 lists nine claims revision 1 got wrong.

PET is diagnostic method development, not a publication product.

Where the selection stands

No configuration is selected, and this deck cannot select one. It compares components; the decision needs one matched comparison of the two *complete* configurations on our unfolding task.

what we have	what it supports	what it does not
a pinned, code-traced inventory of every consequential difference	knowing <i>what</i> differs and which differences are cheap to change	any claim that either configuration unfolds better
component measurements on a synthetic fixture and on GPU cost	cost, and one inconclusive routing result	transfer to the real task — direction and magnitude do not carry
a <i>closure diagnostic</i> recovery of 0.5126 and a frozen injection protocol	a protocol to reuse, and an order of magnitude	a production baseline. Its artifacts are quotable: <code>False</code> and <code>recovery_evaluated</code> is <code>False</code> at the promoted configuration (OI-71)

Provisional engineering choice, stated as such: retain ours — it is frozen, validated, cheaper, and its closure receipts exist. Retention on continuity and cost is not a demonstrated result, and this deck does not upgrade it into one.

What would settle it: slides 16–19.

What revision 1 got wrong

revision 1 said	corrected
"the exact upstream configuration is the Transformer1 preset"	Transformer1 is not a model his paper reports. V1Paper.json lists OmniLearned-small / -small-rw / -medium and puts Transformer-xsmall/small under <code>_disabled_models</code>
his model is $18.9\times$ ours	that is Transformer1. His <i>paper</i> backbone is $58.7\times$ (2,762,550 params); medium is $1,112\times$
the representation is "blocked on a C++ deliverable"	partly false. Cap, overflow, merged counts and discarded energy need no C++ — the full-length vectors are already in the ROOT and the truncation is a Python choice
cited the interface request "§D" for typed objects	§D is <i>residual-energy summary tokens</i> . No filed request for the typed vocabulary exists. §§A–C landed. Owner: Agent A
"our production baseline is 0.5126033"	that is VL100 from a closure whose every artifact is NONQUOTABLE-DIAGNOSTIC.; recovery_evaluated is False at the promoted configuration. δ rested on it
V-PORT checked forward agreement	forward agreement does not show <i>training</i> is equivalent. Now forward, gradient and weight-update
sizing used normal quantiles	at $n \approx 8$ that understates the interval. Now t-based, iterative, sized on the upper bound of the pilot's σ
"one endpoint"	the injection is a tilt in truth muon p_T — a muon kinematic , and the closure is separately silent about the 34 % fold-forward deficit
70 GPU-h as the campaign cost	70 GPU-h is a Tier-A ceiling only. Completing the goal depends on a ratio stage 2 must measure

What is pinned, and what “ours” means

label	what it is	pin
ours, production	pet-fullevent-fps-v1: the estimator the driver stamps	44142e8c (branch pet-direct-token-comparison)
ours, experimental	four-arm representation study: typed descriptors, routing, aggregate overflow. No arm is in the production path.	44142e8c, direct_token_comparison/
ours, proposed	what this deck recommends changing	nothing is built
Gregor's	gregorkrz/minerva-ml	fc9a099d, 2026-07-20, clean tree

- ▶ **fc9a099 is the repo as cloned, not a paper commit.** No release tags; the repo has evolved past arXiv:2604.12364. The paper commit must come from Gregor.
- ▶ Everything attributed to him is the **default** unless marked optional. Verified at fc9a099: use_pid on, use_cond on, include-E-sum on, max_particles 33, **use-max-blobs-and-prongs off** — so his overflow aggregation is optional and not the default.
- ▶ **He is not doing unfolding.** Supervised regression/classification, MC only, no data leg, no per-event weights. Several rows below have no counterpart, and that is a difference in problem, not in quality.

The actual baseline: our production estimator

Data path

- ▶ cloud = **raw non-muon calorimeter clusters** ($E, \text{pos}, z, \text{view}, t$), 5 columns
- ▶ muon **removed from the cloud**; carried as a 13-feature event block, FiLM conditioning
- ▶ cap: stable energy sort, **top 12**, zero-pad — applied **at dump time**, so changing it needs a re-dump
- ▶ overflow **discarded with no trace**: no count, no summed energy, no flag
- ▶ fixed physical divisors; event features z-normalized with *frozen* reco-MC pass_reco statistics
- ▶ azimuth as $(\cos \phi, \sin \phi)$ everywhere

Model and training

- ▶ PET: encoder $\rightarrow$ **KNN local aggregation** ($K = 3$ over (pos, z)) $\rightarrow$ 2 blocks, 2 heads, width 32, LayerScale $\rightarrow$ class token + FiLM head
- ▶ **47,041 trainable parameters**; 93,954 per OmniFold iteration
- ▶ weighted BCE; Adam 10^{-4} annealed to 10^{-5} , realized LR *asserted* at runtime; batch 512; niter 3, epochs 8; 2 M subsample
- ▶ MultiFold; push weight $w = \exp(\text{logit})$ in logit space
- ▶ background: negweight-refined, fail-closed
- ▶ seed 42 frozen — this is what makes C_{stat} statistical

`train_fullevent_nominal.py:403-407, 47-71; fullevent_fps_dataloader.py; dump_pointcloud_inputs.py:90-112; omnifold_nn/omnifold/{net,omnifold}.py; receipts/model-capacity.json`

Gregor's implementation — the repo baseline, **not his paper model**

Transformer1 is under `_disabled_models` in `V1Paper.json`. The paper's lineup is OmniLearned — next slide. This page is retained because the interim inventory was built against it.

Data path

- ▶ cloud = **reconstructed objects**: muon, ≤ 2 photons, blobs, prongs — 10 columns each
- ▶ η , ϕ , $\log p_T$, $\log E$, **PID code** (8-class learned embedding), dE/dx , x, y, z, t (all $/10^4$)
- ▶ muon is **a token like any other** (PID 0)
- ▶ caps in series: 150 at preprocessing (energy-sorted *only* if exceeded), then **first 33 rows at batch time, no re-sort**
- ▶ absent fields **zero-filled, no validity mask**
- ▶ 16 global features: recoil calorimetry, passive material, Michel, $\gamma\gamma$ mass, prong count, **6 per-PID log energy sums**

Model and training

- ▶ PointGlobalMixedViT: mixed encoder $\rightarrow$ **Fourier positional MLP over** $(\eta, \phi) \rightarrow 4$ pre-norm blocks, 8 heads, $d_{\text{model}} = 128 \rightarrow$ CLS fused with EVT token
- ▶ **890,130 parameters** — $18.9\times$ **ours**, but **not a paper model**
- ▶ Huber on $\log 1p$; **AdamW**, wd 0.01, **1000-step warmup then cosine, grad clip 1.0**; batch 2048; 100k–500k steps
- ▶ optional: OmniLearned PET2, HyperScale, BERT, pretrained init
- ▶ **pretrained checkpoints are unreachable** — HyperScale repo 404, NERSC URLs time out (audit B_provenance)

vit.py:226-373; train.py:465-756, 2328-2358; preprocessing.py:257-583; jobs/submit_train_jobs.py:150-151;
gregor-audit/jobs/A_dataset.out.md

Which configuration is “Gregor’s”? Three, and only one is runnable

Paper: *Cross-Domain Transfer with Particle Physics Foundation Models: From Jets to Neutrino Interactions* (arXiv:2604.12364). The title states the subject — **transfer from a pretrained foundation model** — so the pretrained arm is the contribution and a from-scratch transformer is a baseline.

	id	model	init	params	runnable?
P1	OLS	OmniLearned-small	pretrained pretrain_s	2,762,550	no — R2
P2	OLS_RW	OmniLearned-small	scratch	2,762,550	yes
P3	OLM_FB	OmniLearned-medium, backbone frozen	pretrained pretrain_m	52,294,730	no — R2
B1	Transformer1	PointGlobalMixedViT	scratch	890,130	not a paper model
	ours	PET, 2 blocks, width 32	scratch, seed 42	47,041	—

Checkpoints: best_model_pretrain_{s,m}.pt from portal.nersc.gov/cfs/m4567/checkpoints — CFS project **m4567**, not ours; **verified to time out**, and the code **silently attempts a download when the file is absent**.

Backbone as his code builds it (train.py:1085-1105, not class defaults): input_dim 4, add_dim 5, pid_dim 8, cond_dim **16**, num_coord 2, K=**10** (hardcoded, not the PET2 default 15); small = 8+2 transformers, 8 heads, base_dim 128, mlp_ratio 2.

All three paper arms carry **-zero-cond-feature 2**, zeroing the log-hadron-recoil global — his configuration *deliberately removes a hadronic summary*. **P2 alone answers “is his architecture better”**; it does not answer **“is his paper method better”**, because the paper’s claim is about transfer.

Comparison I — representation

	ours	Gregor's	consequence	recommendation
objects	raw non-muon calorimeter clusters; muon distinguished	reconstructed objects; muon is a token	mean 74.6/102.6 clusters vs 14.0/15.6 objects: $5.3\times/6.6\times$ fewer tokens. Whether they carry the same information is unmeasured	borrow the vocabulary; keep the distinguished muon
token features	5 cols, no type code; mask = ($E=0$)	10 cols incl. direction, dE/dx , time, PID embedding	more physics per object; but zero-fill hides absence, and padding shares PID index 0 with muon (masked, so latent)	borrow features + embedding; reserve index 0 for padding
transforms	physical divisors; frozen MC z-norm; $(\cos \phi, \sin \phi)$	logs $+/-10^4$; LayerNorm inside encoder; raw ϕ	raw ϕ is discontinuous at the wrap; we fixed this in July	keep ours; recommend the ϕ fix to him
event globals	13, all muon/vertex	16, recoil calorimetry + 6 per-PID energy sums	his sums are a fixed-width channel that survives a cap — but he computes them <i>after</i> truncation	borrow the sums, computed <i>before</i> any cap
caps and overflow	top 12 by energy, rest discarded silently	150 then first 33 unsorted; aggregate token <i>optional</i> , off	our cap binds 64%/84% vs his 9–17%/11–27%; his ordering is worse	borrow aggregation; keep energy ranking

Comparison II — model, training, integration

	ours	Gregor's	consequence	recommendation
routing	one family, so no question in production; pooled vs individual measured experimentally	individual, unconditionally	no accuracy difference detected; individual costs $1.24\times$ at the operating point, $1.88\times$ in the tail	keep pooling as a practical default
architecture	2 blocks, 2 heads, width 32, KNN in (pos, z) ; 47k params	4 blocks, 8 heads, width 128, Fourier PE in (η, ϕ) ; 890k params	$18.9\times$ to Transformer1; 58.7$\times$ to his PAPER backbone . Ours is retrained ~ 130 times per campaign, his once	keep architecture; treat capacity as open
init	random, seed 42 frozen (this makes C_{stat} statistical)	random; optional pretrained OmniLearned	checkpoints unreachable : 404 and timeouts	keep; ask Gregor for the weights and hashes
training	Adam $10^{-4} \rightarrow 10^{-5}$ step anneal; no wd, no clip	AdamW, wd 0.01, warmup + cosine, clip 1.0	<i>judgement, unmeasured on our estimator</i> : his schedule is better conditioned, and free. wd biases a likelihood-ratio classifier ($w = e^{\text{logit}}$)	borrow warmup/cosine/clip; not weight decay
OmniFold	full MultiFold, logit-space ratio, negweight	no counterpart — MC only, no weights, no flags	his <i>dataset</i> cannot be an OmniFold input;	keep ours in full

The extraction contract — and the blocker is alignment, not physics

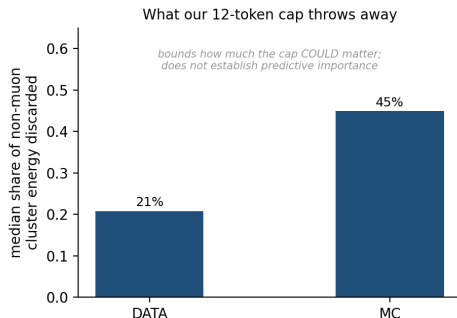
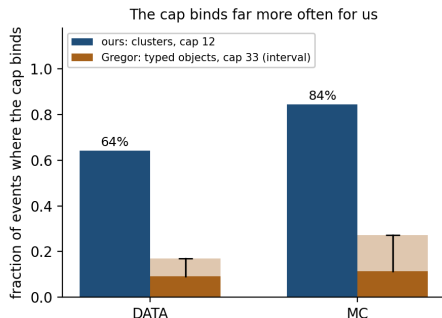
Gregor field	our branch	tuple	ROOT	npz	needed
muon 4-vec, ϕ , q/p , MINOS; vertex; view/time	mu_reco_*, vtx_reco_*, part_reco_view/time	✓	✓	✓	nothing
cap, overflow aggregate, merged count, discarded energy typed objects — photon, blob, prong: 4-mom, position, time, PID, dE/dx	full-length part_reco_* vectors	✓	✓ full	trunc. 12	dump change. NO C++
16 event globals	gamma{1,2}_*, MasterAnaDev_Blob*, prong_part_*, prong_dEdXMean fuzz, iso-blob, hadron recoil, passive, Michel, ...	✓ all authorized	—	—	export
		✓ not authorized	—	—	authorize + export

Every typed field already exists in the **AnaTuple** and is already inside the **A1-authorized branch set** — A1 read all of them on the real tuples. And the **cap** is **ours**: the dump binds full-length vectors and `_pad_tokens` truncates them, exactly as the interface request says.

The blocker is the join key. The **AnaTuple** carries `ev_run/ev_subrun/ev_gate`; **the npz carries none** — rows are positional. So a Python-side join is impossible *not because data is missing but because the key is discarded*. Two routes, both requests to **Agent A** (who owns the C++; the request says do not edit it): **R-1 emit three scalar key branches** and join in Python, keeping the representation ours — or **R-2 emit ~20 typed vector branches**. **R-1 is an order of magnitude smaller ask**, at the price of having to *prove* the join.

Any route must preserve the FPS gate, `pass_reco/pass_truth` with native truth-only misses *kept*, `−9999` sentinels, RAW weights, the negweight background inventory, CLM-007's fail-closed data leg, the identity hashes, and a **per-event order proof** — row-count alignment alone is already flagged as insufficient.

The measured problem: our cap binds, and it binds hard



Unselected, unweighted tuple entries: 17,930 data (1B) + 186,439 MC (1A). Not the selected population.

Same two tuple files, both vocabularies. Our 12-token cluster cap binds in **64.1 % (data) / 84.4 % (MC)** of events and discards a median **20.8 % / 44.9 %** of non-muon cluster energy. A 33-token typed-object cap binds in **9.2–16.9 % / 11.4–27.2 %** — an interval, because the receipt holds marginal, not joint, family multiplicities.

Three things this does not show. It does not show the discarded clusters carry predictive information. It does not predict a closure improvement — **no closure measurement of aggregation exists**. And these are **unselected, unweighted** tuple entries, so they do not determine selected-population behaviour.

`receipts/typed-object-budget.json; source-multiplicity.json (job 58470099)`

Measured accuracy and stability: routing is inconclusive

Eight paired seeds, four-object synthetic fixture,
information content / model size / budget / init / seeds
held identical between arms.

paired improvement, individual over pooled	
median	+9.7 %
mean	+0.8 %
seed-to-seed sd	25.8 points
95 % interval on the mean	$[-20.7, +22.4]$ %
paired p	0.50
favourable seeds	6 of 8

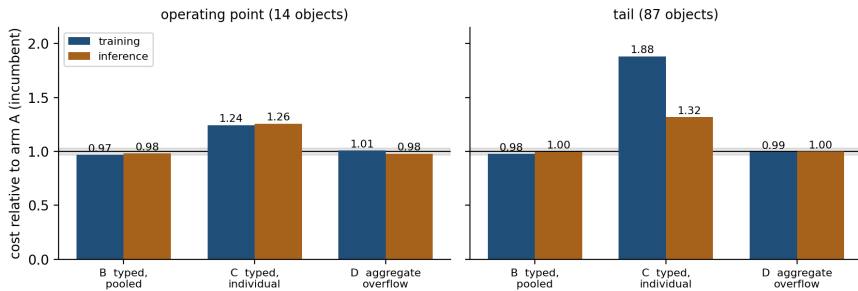
Frozen decision: NO_PASS, 142 of 146 checks holding.
Two failures *were* the inconclusive result; separately
shuffle-71 missed two projection checks marginally
(0.0119 vs 0.01; absolute errors well inside 0.05).

What each number measures —

- ▶ the interval is on the **mean paired injected-improvement percentage across training seeds on a synthetic fixture**. Not on production closure, not on real data.
- ▶ $p = 0.50$ is a paired test **across training seeds**. It measures seed scatter. **Test-sample uncertainty was not separately estimated**.
- ▶ 25.8 points is a **planning assumption** carried from a different fixture, two arms and a different denominator — **not measured power for the endpoint now specified**.

And: in the fixture every generic slot is noise while in production the cluster cloud carries most of the information. Neither direction nor magnitude of a synthetic effect transfers automatically.

Measured cost: what each change actually buys and spends



Grey band: $\pm 3.3\%$, the resolution of this measurement, estimated from the arm-A/arm-B null contrast. A bar inside it is not a difference.

One A100, TF32 off, determinism on, float32. Cost is cost: a cheaper arm is not a better one.

Aggregate overflow is *not* free — it is indistinguishable from parity at this measurement's resolution. Arm B is computationally identical to arm A in the pooled configuration and still reads $0.97\times$, so the floor is about $\pm 3.3\%$. The non-compute cost is real: a dump regeneration, a schema field, a re-validation, and a new estimator fingerprint.

Individual routing is outside that band and grows with multiplicity ($1.24\times \rightarrow 1.88\times$ training). Pooling is nearly flat. Our campaign multiplier is ~ 130 trains, so per-train cost is multiplied, not absorbed.

`cost-half.json` (one A100, TF32 off, determinism on, float32); `inference-benchmark.json`

Component inventory — all provisional, none selected

borrow (candidate)	why	cost	prerequisite
per-family log energy sums, over the <i>whole</i> event	cap-proof fixed-width channel	est. ≈ 0	fingerprint change; absent from the dump
aggregate overflow, energy-weighted merge	addresses a measured 21–45 % energy loss in most events	<i>measured</i> parity to $\pm 3.3\%$	dump regeneration; no closure evidence yet
richer per-token features + type embedding, index 0 reserved	more physics per token, fewer tokens	unmeasured	absent from the dump
warmup + cosine, gradient clipping	better-conditioned schedule	est. ≈ 0	Gate-4 fingerprint; closure re-run
one capacity step	$58.7\times$ gap to his paper backbone is a live hypothesis	multiplies through the UQ campaign	price against the campaign, not one train

These rows are component judgements, not a selection, and the single-factor experiments they imply are **RETIRED** — the four live items interact, so they are compared as complete configurations instead (slides 16–19). The rows survive as the record of what differs and as the source of the tuning grids.

Back to Gregor: encode ϕ periodically; reserve embedding index 0 for padding; map raw prong PID 9 (it currently raises); compute the per-PID energy sums before truncation.

Limitations — what these numbers do not say

1. “Negligible measured overhead” is not “free”. Aggregate overflow is inside the $\pm 3.3\%$ resolution of the cost measurement, and it carries a dump regeneration, a schema change, a re-validation and a new fingerprint.
2. Discarded energy does not establish closure improvement. 21–45 % bounds how much the cap *could* matter. No closure measurement of aggregation exists on either side.
3. Validation is incomplete: 10 of 13 widths released. Three widths failed the finite-difference check V5 and were not released. Diagnosis: the *criterion* was wrong, not the gradient — error falls as h^2 (1.8×10^{-2} at the frozen $h=10^{-2}$, 3.1×10^{-4} at 10^{-3}) then rises, the signature of truncation giving way to cancellation. **The criterion was not relaxed after the failures.**
4. Unselected source measurements do not determine selected behaviour. Two files, unweighted, no production selection, no POT weighting.
5. Every statistical quantity is labelled by what it measures — seed scatter, not test-sample uncertainty; a synthetic fixture, not production; a planning assumption, not measured power.
6. The pin is the repo, not the paper. fc9a099 may not be what produced arXiv:2604.12364, and the paper’s pretrained checkpoints are unreachable, so its central transfer claim is untestable here.
7. Our own recovery number is a closure diagnostic, not a baseline. 0I-71: the artifacts are quotable: False, recovery_evaluated is False at the promoted configuration, and the closure **does not exercise the 34 % fold-forward deficit** — it is silent about it. Nothing here discharges 0I-71.

One engine; a port checked three ways; a *hadronic* endpoint

One engine. Both arms inside our TensorFlow MultiFold. The prior campaign declined a cross-framework comparison because it would “confound framework, representation, and training engine”; one engine removes two confounds.

Port checks — forward is not enough. Forward agreement does not show *training* is equivalent: an optimizer or initializer difference reproduces outputs and diverges after one step. So P-1 parameter counts and shapes **exactly equal**; P-2 **forward** $\leq 1e-5$ in float64 over 1,024 fixed inputs including fully-masked and single-token rows, after transferring his weights in; P-3 **gradient** agreement w.r.t. every parameter, tolerance set by a *measured* step-size sweep rather than asserted; P-4 **weight-update** after one optimizer step, reported per tensor; P-5 masked slots unable to reach the output; P-6 repeatability and reload bitwise. **Any failure stops the campaign.**

The endpoint had the wrong variable. The predeclared powered closure injects a tilt in **truth muon** p_T . Our muon block is 13 features conditioning the transformer by FiLM, so that tilt can be transported almost entirely by the muon block — **both arms can score well while differing arbitrarily in the hadronic cloud, which is the only thing Tier B varies.** Using it alone would be choosing an endpoint insensitive to the treatment.

V1, the primary endpoint: a clipped exponential tilt in truth E_{avail} . Already in the npz — `truth_scalars` col 2 from `MC_eavail` — so **no new export**, and available hadronic energy is exactly what a hadronic representation must encode. Protocol mirrors the p_T injection exactly (clipped, rate-preserving, truth-passing rows, disjoint 2 M halves at split seed 7) so the two differ in **one** variable. **V3** keeps the p_T injection unchanged as a muon-side control. **Joseph ratifies the variable** — it decides what the comparison is sensitive to.

V2 is a prerequisite, not an extra. 0I-71: the closure’s fold-forward is ≈ 1.011 against the nominal’s **0.736746**, so **it does not exercise the $\sim 34\%$ deficit — it is *silent* about a live failure mode.** 0I-125: the driver has **no fold-forward computation** at all.

Three separate questions, and a switching threshold

1. **Absolute adequacy**, asked of each arm alone: does it clear $\text{recovery} \geq f \times \text{ceiling}$ on V1 and V3 with V2–V9 passing? **An arm failing this is not recommendable regardless of the comparison** — and two inadequate arms make “ours is non-inferior” a statement about two unusable configurations.

2. **Non-inferiority**: $\delta = 0.017$ recovery points. **Revision 1 justified δ from the margin over the adopted criterion. That rested on VL100, a quotable: False number at a configuration where recovery was never evaluated — withdrawn.** The replacement is a *decision anchor*: adopting the LR anneal knowingly paid -0.0342 of recovery to buy a 33-point normalization repair, and Joseph adopted it. So ~ 0.034 of shape loss was acceptable *when something was bought with it*. Non-inferiority buys nothing, so **δ is half that**: at most half the loss the campaign accepted for a real benefit. It is $\sim 2.8\%$ of the reference ceiling, far above the closure’s repeat spread ($1.2e-3$) and far below the $\sim 19\%$ by which the estimator already sits under the dilution ideal. **Provisional until Joseph ratifies it.**

3. **Superiority, and when we keep ours anyway**: $\delta_{\text{switch}} = 0.02$. Adoption costs a new estimator fingerprint (invalidating every v1-keyed artifact including the Gate-2 target), full re-validation of every covariance component, a **$58.7\times$** model through ~ 130 campaign trains, and for P1/P3 a permanent dependency on a checkpoint we cannot obtain. $\delta_{\text{switch}} > \delta$ is **deliberate**: retaining a validated incumbent is cheap, replacing it is not.

outcome	conclusion
both adequate, CI lower bound $> -\delta$	Recommend OURS
both adequate, his advantage CI $> \delta_{\text{switch}}$	Recommend HIS
both adequate, his advantage demonstrated but $< \delta_{\text{switch}}$	Retain OURS — and say he scored better , naming the switching costs. Never “no difference”
CI contains $-\delta$	Inconclusive — no selection ; incumbent is a <i>provisional engineering choice</i>
either arm inadequate	that arm not recommendable; if both, report the configurations as inadequate

Tier-A ceiling $\neq$ the cost of completing the goal

What decides completion cost is one ratio r : his backbone's per-evaluation cost relative to ours, in our engine, on GPU. Revision 1 implied $r \approx 1$; the $58.7\times$ parameter ratio suggests $\sim 80\times$. Both are wrong, and an indicative measurement says why:

	batch	ours	PET2-small	ratio	PET2-medium	ratio
	64	175.2 ms	278.2 ms	$1.6\times$	1338.4 ms	$7.6\times$
512 (production)		386.1 ms	2255.3 ms	$5.8\times$	15792.6 ms	$40.9\times$

The ratio moves $3.6\times$ across an $8\times$ batch change, so the small-batch figure measured framework overhead — the script says so in its own verdict field rather than letting a reader take $1.6\times$ as the answer. CPU and cross-framework, so not the GPU ratio. What it establishes: the parameter ratio is a bad proxy; at production batch his small backbone is single-digit multiples, so $r \approx 6$ is the working expectation and completion is plausibly affordable; medium is a different cost class. At $r = 6$ a complete 8-seed comparison is ≈ 84 GPU-h; at $r = 20$, 252 — the whole remaining ceiling; at $r = 41$ (medium), 504, out of reach. Stage 2 is therefore a go/no-go on the goal, not a formality. Until r is measured on GPU in one framework, no total is quotable and this deck quotes none — but plan against $r \approx 6$, never against the parameter ratio.

Stages, and what stops it

stage		GPU-h	gate
0	specify V1; draft freeze; file R-1 with Agent A; send R1–R3	0	Joseph ratifies V1 and δ
1	port; P-1... P-6	0.5	all P-* pass
2	cost calibration — measure r	3	go/no-go on the goal
3	tuning, 4 trials per arm	12	both arms frozen
4	variance pilot, 4 paired seeds	12	upper-bound σ admits δ
5	Tier-A comparison, 8 paired seeds contingency	24	—
		18	
Tier-A ceiling		≤ 70	of 290; 16.0 consumed
6+	Tier B — the complete comparison	$8c(1+r)$	R-1 landed, R2 obtained, r known, new authorization

Stops: a port check fails; r puts completion outside the allocation (**say so — do not raise the ceiling, narrow the endpoint or widen δ**); either arm fails absolute adequacy; the pilot's upper-bound σ cannot resolve δ ; limits exhausted, with repairs counting as retries and no automatic relaunch; or the endpoint, δ , δ_{switch} , the decision rule or the samples would have to change — which is a new approval, not an amendment.

Frozen before any comparative number exists, digest-verified: both configurations, the forced adaptations, the V1 protocol, the sample, δ , δ_{switch} , the fairness definition and the rule verbatim. **Not relaxed after failures.**

What the final deck will lead with

Once stage 5 returns, the deck is restructured so its first three slides are:

1. **the selected configuration**, named in one line, with the word “selected” reserved for a case the decision rule actually licenses — and the words “provisional engineering choice” used instead when it does not;
2. **the matched evidence**: the paired recovery difference and its 95 % interval against δ , the signed-response / scatter split that attributes the difference, and the safeguard table;
3. **the conditions under which the conclusion holds** — Tier A only, so architecture and training recipe and *not* representation; one endpoint; $\delta = 0.018$; `fc9a099d` not established as the paper commit; pretrained initialization untested because unreachable; recovery measured against a reference ceiling that is not a proven bound.

The present inventory becomes the appendix. Two sentences are pre-committed now, so neither can be softened later:

If his configuration wins by more than δ , the deck recommends his and says so in the title.

If the interval is wide, the deck says the comparison failed to decide and names the incumbent as retained on cost and continuity — not as validated.

Open regardless of the outcome: the representation (Tier B, blocked on the dump), his published configuration (R1), and pretrained initialization (R2). **Questions for Gregor, not compute**: R1 paper commit, R2 checkpoints and hashes, R3 his E_{avail} definition. **Nothing has been sent to anyone.**

Appendix A — claim-to-evidence index

claim	evidence
cap binds 64.1 % / 84.4 %; median 20.8 % / 44.9 % energy discarded	direct_token_comparison/local_validation/20260918-source/source-multiplicity.json (job 58470099; 17,930 data + 186,439 MC entries)
typed cap binds 9.2–16.9 % / 11.4–27.2 %; 5.3×/6.6× fewer tokens	receipts/typed-object-budget.json (derived; 8 tests)
47,041 vs 890,130 parameters	receipts/model-capacity.json (both models instantiated)
routing: median +9.7 %, [−20.7, +22.4] %, $p = 0.50$, NO_PASS 142/146	local_validation/20260917-matrix/matrix-summary.json
inference 1.283×, all 5 checks pass, $CV \leq 2\%$	local_validation/20260917-inference/inference-benchmark.json
GPU cost by arm; A/D parity; $\pm 3.3\%$ floor	local_validation/20260918-a3-cost/cost-half.json
10 of 13 widths released, no hard stop	local_validation/20260918-tailval/validation-final.json; criteria sha256 abccd88b...
Gregor token format, PID scheme, caps, globals	gregor-audit/jobs/A_dataset.out.md (line-cited to fc9a099)
checkpoints unreachable; OmniLearned MIT;	gregor-audit/jobs/B_provenance.out.md
dataset CC-BY-4.0	
our production configuration	FULL_EVENT_FEATURE_CONTRACT.md; train_fullevent_nominal.py:47–71, 403–407

Full index: ../CLAIM_EVIDENCE_INDEX-20260918.md. Full inventory: ../CONFIGURATION_COMPARISON-20260918.md.

Appendix A2 — anchors for the matched comparison

claim	evidence
closure-diagnostic recovery 0.5126033 (VL100) against the adopted criterion 0.494582 . NOT a production baseline: every artifact is NONQUOTABLE-DIAGNOSTIC. with quotable: False, and recovery_evaluated is False at the promoted configuration	job 56552326, finalized 56562169; CLAIM-CLM-012.md (viii); docs/OPEN_ITEMS.md OI-71, WAITING-USER, G4 alone surviving
the closure's fold-forward is ≈ 1.011 vs the nominal's 0.736746 — it does NOT exercise the $\sim 34\%$ deficit and is SILENT about it	OI-71(4)(e); OI-125 (the closure driver has no fold-forward computation at all)
paper lineup: OmniLearned-small / -small-rw / -medium; Transformer1 disabled	plot_configs/V1Paper.json; submit_train_jobs.py:155-169
paper backbone capacity 2,762,550 ($58.7\times$ ours); medium 52,294,730	receipts/model-capacity.json
step-cost ratio $5.8\times$ at batch 512, $1.6\times$ at batch 64 — overhead-dominated, not a scaling estimate	receipts/step-cost-scale.json (CPU, cross-framework)
truth E_{avail} is already in the npz, truth_scalars col 2	dump_pointcloud_inputs.py:79 (MC_eavail)
the npz carries no event key ; the AnaTuple carries ev_run/ev_subrun/ev_gate	dump_pointcloud_inputs.py:190-235; typed_descriptor_source_smoke.py:86

Appendix A3 — anchors, continued

claim	evidence
closure repeat spread 1.226e-3 — same-configuration, <i>not seed scatter, so it sizes nothing</i>	job 56611837; CLAIM-CLM-012.md (ix)
97.8 % of the baseline's gap to the ceiling is scatter; signed bias -0.0019	FINDINGS-ARCHIVE-2026-08.md BEN-038
the ceiling 0.618228 is a <i>reference curve, not a proven bound</i> (a band was measured at $E_w[r] = 1.0333$)	CLAIM-CLM-012.md caveat (i); BEN-038
the ordinary closure has <i>structural zero power</i> injection protocol predeclared 2026-08-05, reused unchanged	AUDIT-FINDINGS-20260728.md; closure_fullevent_fps.py:4-9 closure_powered_truth_reweight.py:10-33
evaluation sample: 49.15 M signal / 4.12 M data / 0.56 M background rows, sha256 fa6b3463..., 9.90 GB	g2_fullevent/input/G2_FPS_MEFHC_P12_RECEIPT.json, status PASS, job 56120687
<i>his representation is absent from every inventory</i>	dump_pointcloud_inputs.py:190-235 at 44142e8c (re-measured); GREGOR_PET2_OMNIFOLD_ASSESSMENT.md:249-255 at b65f9ff2
seed-scatter prior $\sigma \approx 0.008$	CLAIM-CLM-010.md: 48 seeds, mean 0.014256, sd 0.008023 — <i>an adjacent statistic, a planning assumption only</i>
a PyTorch OmniFold backend exists (10.9 k lines) but its model is an <i>independent</i> reimplementation, not his	nd-unfolding/pet2_torch/ at b65f9ff2; model.py docstring

Appendix B — the V5 diagnosis, in full

Three widths — (0, 64, 2) and (2, 160, 12) for the individual arm, (2, 12, 4) for both — failed the finite-difference check, and only that check. My first hypothesis (a small denominator) was **wrong**: the failing coordinates have $|\nabla| \approx 0.0215$. Sweeping the step at the worst coordinate:

h	numeric derivative	relative error
3×10^{-2}	-0.01787	1.69×10^{-1}
1×10^{-2} (<i>frozen</i>)	-0.02111	1.84×10^{-2}
3×10^{-3}	-0.02147	1.67×10^{-3}
1×10^{-3}	-0.021497	3.14×10^{-4}
3×10^{-4}	-0.021531	1.30×10^{-3}

Error falls as h^2 and then rises: truncation giving way to cancellation. The analytic gradient is correct; the frozen *global* step was too large at those coordinates, and the optimal step is per-coordinate. The plateau check nearly caught it (16.8% spread against a 20% tolerance).

I did not relax the criterion. Releasing those widths requires an amended step rule. Until then the execution path is validated for 10 of 13 widths, which is **incomplete**, and the variance pilot's precondition is unmet.

Appendix C — what the campaign cost, and what it did not authorize

item	GPU device-hours	source
representation matrix (24 paired jobs)	13.27	matrix-summary.json
A3 geometry probe + width gate (5 subs)	0.43	GO_NO_GO-20260918.md §7
tail validation (4 of 4 submissions)	0.77 of 1.5 authorized	validation-final.json
inference benchmark + matrix closeout	<i>not itemized here</i>	—
A1 source characterization	(CPU: 2.0 core-h)	source-multiplicity.json
campaign total	16.0 of 290	RECOMMENDATION-20260918.md

The first three rows are itemized from their own receipts; the total is carried from the campaign recommendation. They do not sum, because the inference benchmark (including one 35-minute timeout that measured nothing) and the matrix closeout are inside the total but are not separately itemized in any receipt I can cite. Stated as a gap rather than closed by arithmetic.

Not launched, and now retired: the *synthetic* four-arm pilot (≤ 30 GPU-h, precondition unmet) and its Stage 2 (≤ 130 GPU-h). **Do not confuse that pilot with stage 4 of the matched comparison** — different sample, different endpoint, different purpose. **Requested instead:** ≤ 70 GPU-h for stages 1-5 of slides 16-19.

Not authorized by anything in this deck: publication adoption, any covariance or systematic, a central-value change, a Gate-6 action, or the construction of C_{ML} . Typed objects and aggregate overflow remain *prospective* representation improvements, exactly as the analysis note already says.

Not done: nothing in this deck has been sent to anyone.